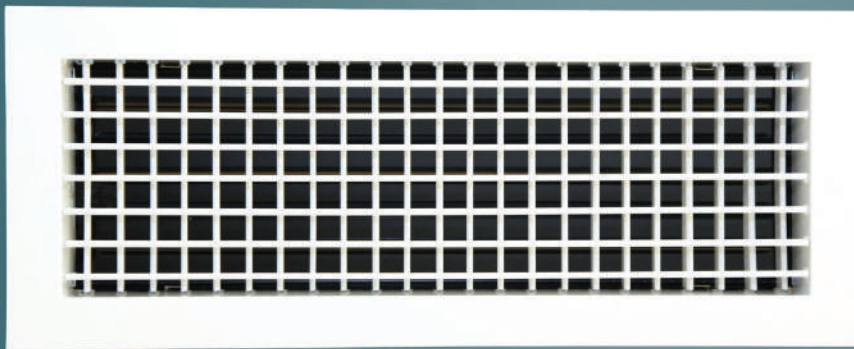




TECNALCO

CHAPTER 2



GRILLES AND REGISTERS

GRILLES AND REGISTERS

CHAPTER-2

GRILLES AND REGISTERS

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SUPPLY, RETURN, EXTRACT GRILLES AND REGISTERS

TECNALCO grilles and registers provide a high standard of quality, solid construction and quiet, controlled performance.

Features:

- Extruded aluminium blades with aerofoil cross section for minimum turbulence.
- The horizontal and vertical vanes are provided with nylon bushes for rattle free operation.
- Borders are one piece construction in size through 1200 mm x 1200 mm, for smooth appearance and easy installation.
- Where grilles and registers are to be butted together in multiples, joining strips are provided in proper lengths at no extra cost.
- Counter sunk screw holes accommodate metal screws, heads flush with surface of border.
- Registers have opposed blade dampers for minimum disturbance of air stream.
- Auxiliary frames available on request for neat and separable mounting in plaster and other wall materials.
- Sponge foam gasket provided on request.

Types available:

Grille Model	H	V	D	F
RG	RG - H RGH - 45°	RG - V	RG - HD RG - VD RG - FD*	RG - FA* RG - F - Fx*
SG	SG - H	SG - V	SG - HD SG - VD	-
ECG**	-	-	ECG - D	ECG - F*
PG**	-	-	PG - D	-

- H - Front Horizontal
- V - Front Vertical
- D - with Damper
- F - with Filter

*Also available with double frame

**Models without accessories

Standard sizes available:

Any combination of L x W = N

L	W
150	100
200	150
300	200
400	300
500	400
600	500
700	600
800	
900	
1000	
1100	
1200	

N = nominal size = Duct size

A = Neck size of the grille

B = Overall size of the grille

where

$N = L \times W$

$A = (L - 10) \times (W - 10) = x \times y$

$B = (x + 60) \times (y + 60) = (L + 50) \times (W + 50)$

Fw = Frame width = 30 mm

Fd = Frame depth = 27mm / 47 mm

Bp = Pitch of the blade = 19 mm

Dp = Pitch of the damper = 25 mm

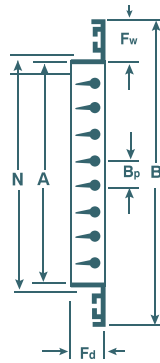
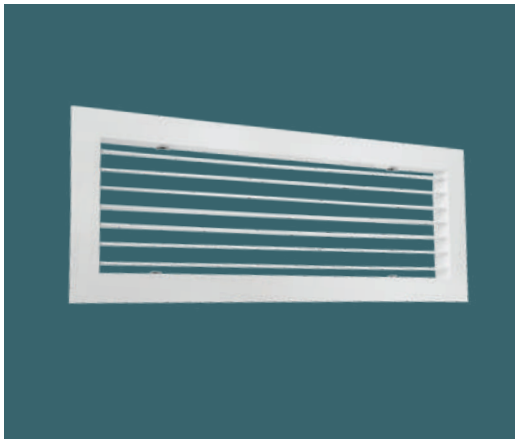
Dd = Damper depth = 36 mm

Ft = Filter thickness = 12.5/25 mm

Fw = 25 mm available on request

Frame thickness = 1.2mm

SUPPLY, RETURN, EXTRACT GRILLES AND REGISTERS



RG-H

TECNALCO model RG-H is a return air grille with one set of horizontal aerofoil blades.

Blades are individually adjustable to any degree of deflection.

Fd = Frame depth = 27 mm

FW = Available size 30mm /25mm

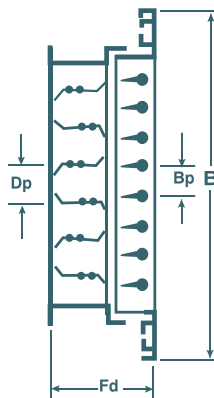
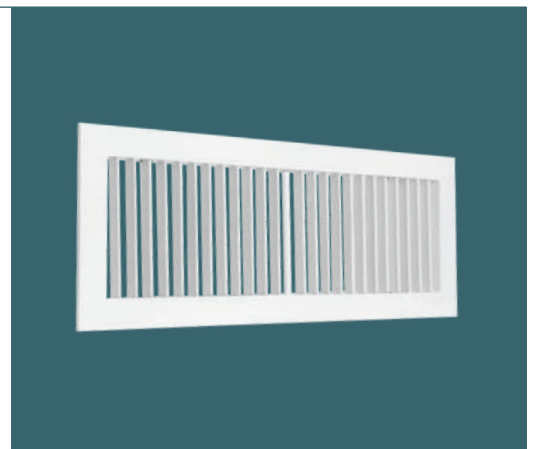
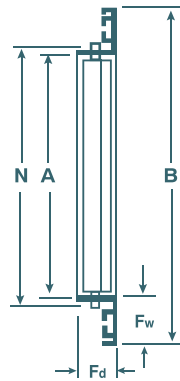
RG-V

TECNALCO model RG-V is a return air grille with one set of vertical aerofoil blades.

Blades are individually adjustable to any degree of deflection.

Fd = Frame depth = 27 mm

FW = Available size 30mm /25mm



RG-HD

TECNALCO model RG-HD is a return/supply air grille with one set of horizontal aerofoil blades and with an opposed blade type volume control damper.

Blades are individually adjustable to any degree of deflection.

Opposed blade damper is screw driver operated from the face of the grille.

Overall grille depth = 27 + 36 mm

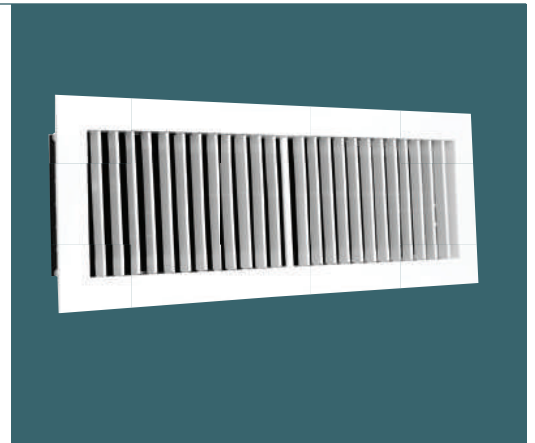
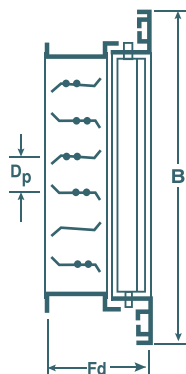
RG-VD

TECNALCO model RG-VD is a return/supply air grille with one set of vertical aerofoil blades and with an opposed blade type volume control damper.

Blades are individually adjustable to any degree of deflection.

Opposed blade damper is screw driver operated from the face of the grille.

Overall grille depth = 27 + 36 mm



SUPPLY, RETURN, EXTRACT GRILLES AND REGISTERS

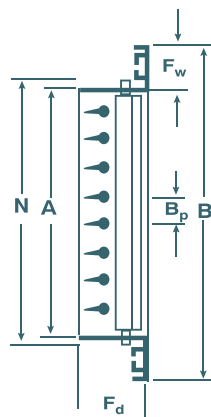
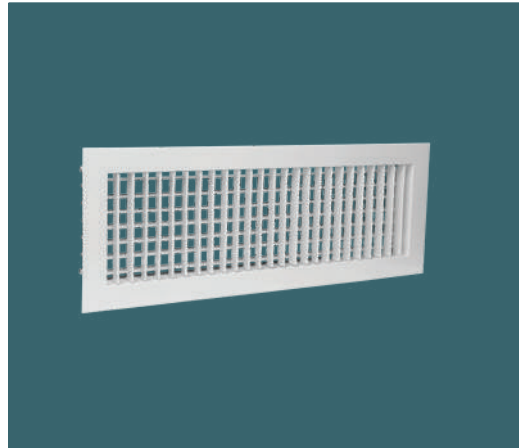
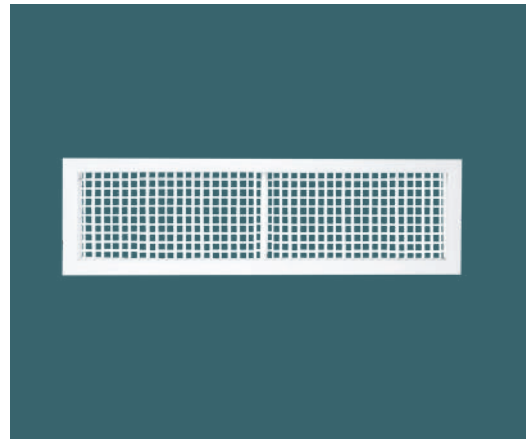
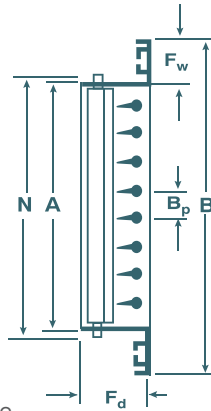
SG-H

TECNALCO model SG-H is a supply/return air grille with two set of individually adjustable (double deflection) aerofoil blades with front horizontal and rear vertical.

All aerofoil blades are individually adjustable to any degree of deflection.

Fd = Frame depth = 47 mm

FW = Available size 30mm /25mm/20mm/15mm/12mm



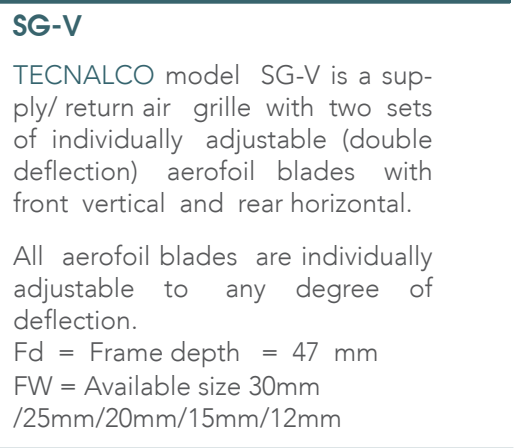
SG-V

TECNALCO model SG-V is a supply/ return air grille with two sets of individually adjustable (double deflection) aerofoil blades with front vertical and rear horizontal.

All aerofoil blades are individually adjustable to any degree of deflection.

Fd = Frame depth = 47 mm

FW = Available size 30mm /25mm/20mm/15mm/12mm



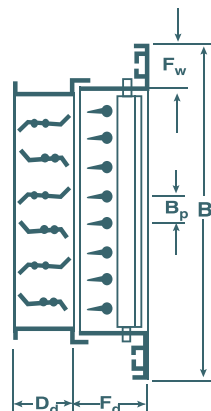
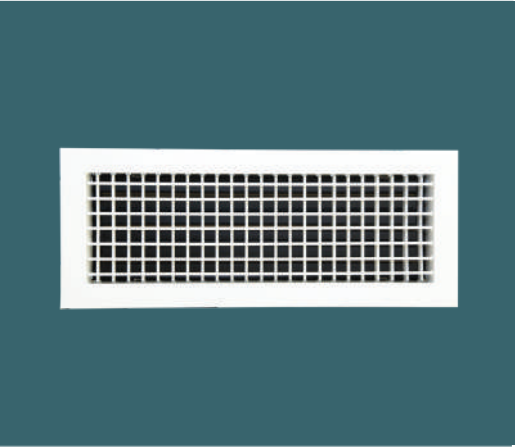
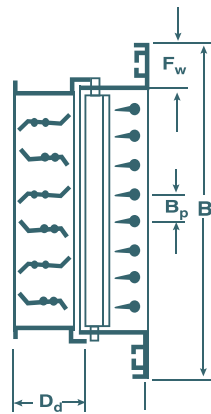
SG-HD

TECNALCO model SG-HD is a supply air grille same as SG-H and with an opposed blade type volume control damper.

Volume control damper can be operated with a key or a screw driver from the face of the grille.

All aerofoil blades are individually adjustable to any degree of deflection.

Overall grille depth = 47 + 36 mm



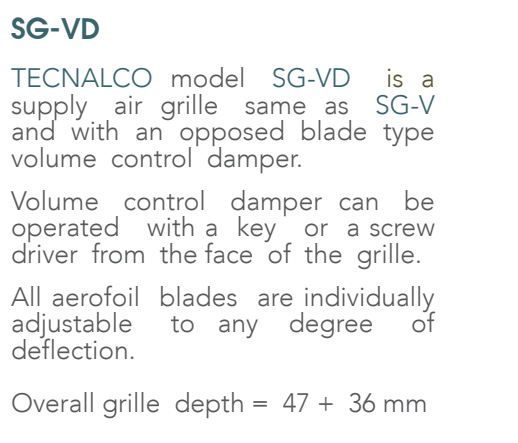
SG-VD

TECNALCO model SG-VD is a supply air grille same as SG-V and with an opposed blade type volume control damper.

Volume control damper can be operated with a key or a screw driver from the face of the grille.

All aerofoil blades are individually adjustable to any degree of deflection.

Overall grille depth = 47 + 36 mm



SUPPLY, RETURN, EXTRACT GRILLES AND REGISTERS

ECG

TECNALCO model ECG is a return air grille of Egg-Crate type with core dimension 12.5 x 12.5 x 0.5mm.

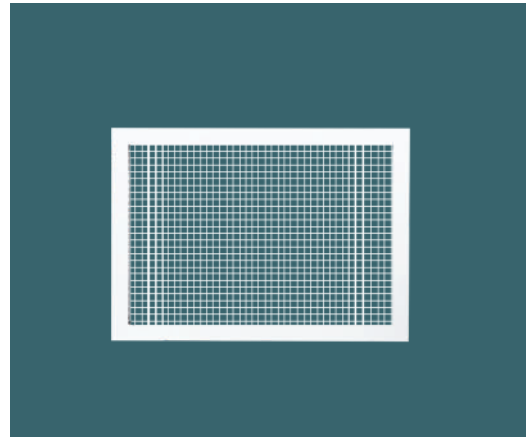
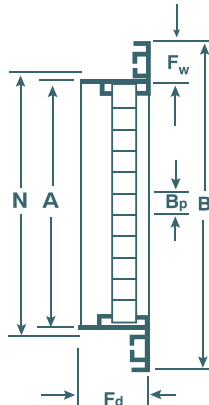
The Egg Crate square core pattern is of aluminium construction providing maximum free area.

Fd = Frame depth = 30 mm

Bp = Core pitch = 12.5 mm

Fw = Frame width = 30mm

Available with fly screen



ECG-D

TECNALCO model ECG-D is a supply/return air grille of Egg Crate type same as ECG model and with an opposed blade type volume control damper.

Volume control damper can be operated with a key or a screw driver from the face of the grille.

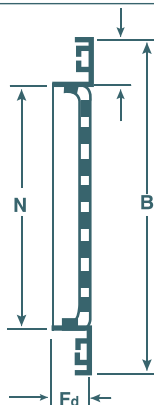
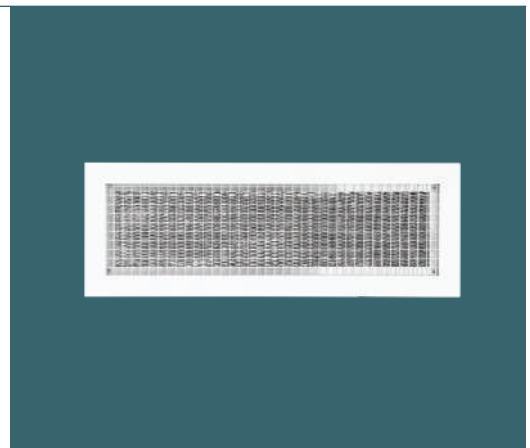
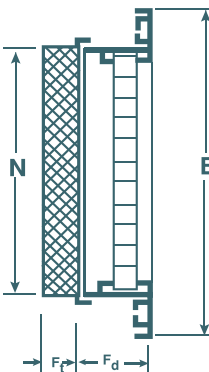
Overall grille depth = 30 + 36 mm
Available with fly screen

ECG-F

TECNALCO model ECG-F is a supply/return air grille same as ECG and with an aluminium filter. Aluminium filter thickness = 12.5/25 mm.

The Egg Crate core pattern is of fixed type.

Overall grille depth
= 30 + 12.5/25 mm



PG

TECNALCO model PG is a perforated return/supply air grille with 3.25 mm dia of perforations at a pitch of 5 mm.

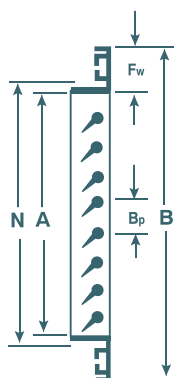
The perforated sheet is of mild steel with powder coated finish.

Also available with opposed blade volume control damper model - PG-D

Fw = Frame width = 30mm

Overall grille depth = 45 mm /
45 + 36 mm

SUPPLY, RETURN, EXTRACT GRILLES AND REGISTERS



RGH- 45°

TECNALCO model RGH- 45° is a return air grille with fixed horizontal aerofoil blades in a 45° angle.

Fd = Frame depth = 30 mm
Bp = Blade pitch = 14 / 17 mm
Fw = Frame width = 30mm

RG-F

TECNALCO model RG-F is same as RGH-45° and with a 12.5 mm thick aluminium filter, used as a Fresh Air Grille.

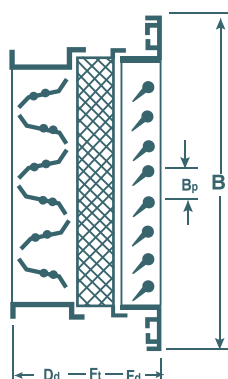
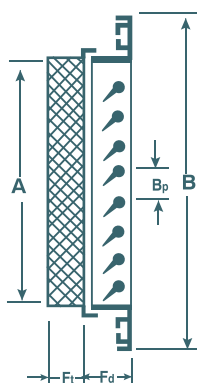
The blades are in either fixed or adjustable pattern.

Overall grille depth = 30 + 12.5 mm
Bp = Blade pitch = 14 / 17 mm

RG - F - A → Adjustable

RG - F - Fx → Fixed

Fw = Available size 30mm/25mm



RG-FD

TECNALCO model RG- FD is same as RG-F with an opposed blade type volume control damper.

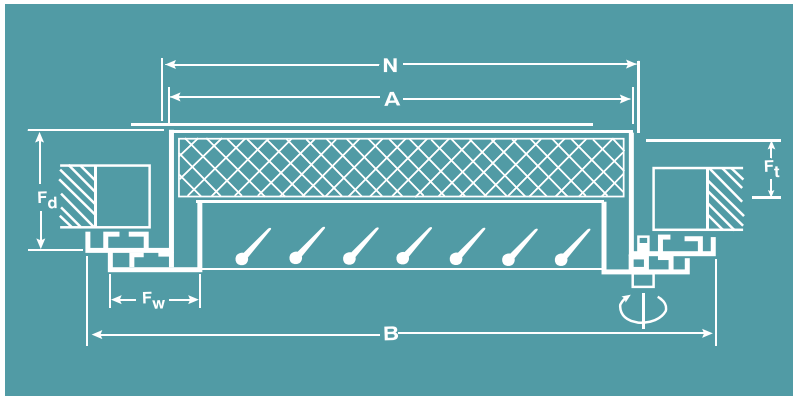
The volume control damper can be operated after removing the filter.

Also available with damper at middle.

Overall grille depth = 27 + 12.5
+ 36 mm

Bp = Blade pitch = 14 / 17 mm
Fw = Available size 30mm/25mm

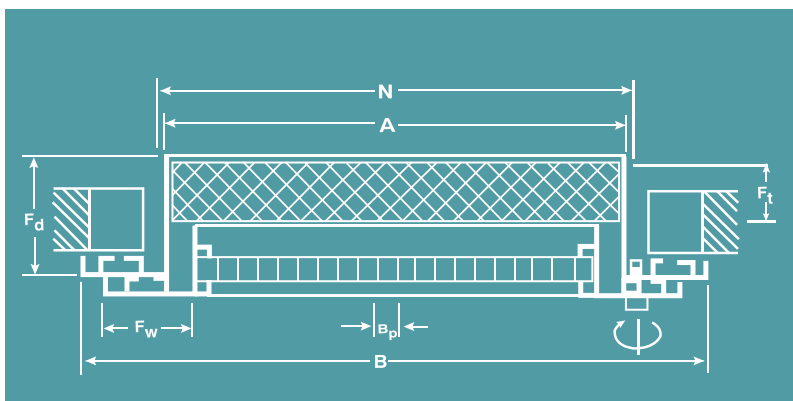
SUPPLY, RETURN, EXTRACT GRILLES AND REGISTERS



RG-F WITH DOUBLE FRAME*

TECNALCO model RG-F with Double Frame is a Fresh air grille with aluminium filter. The filter is detachable and washable type.

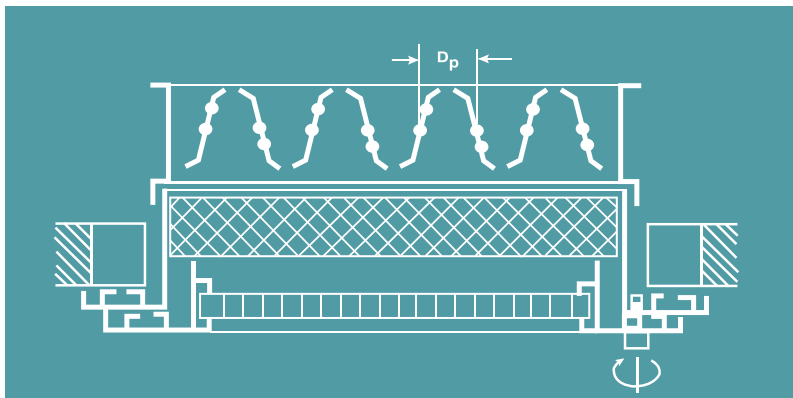
Internal External
Frame depth = 47 mm 27 mm



ECG-F WITH DOUBLE FRAME*

TECNALCO model ECG-F with Double Frame is a Fresh Air Egg Crate Grille with aluminium filter.

Internal External
Frame depth = 47 mm 27 mm

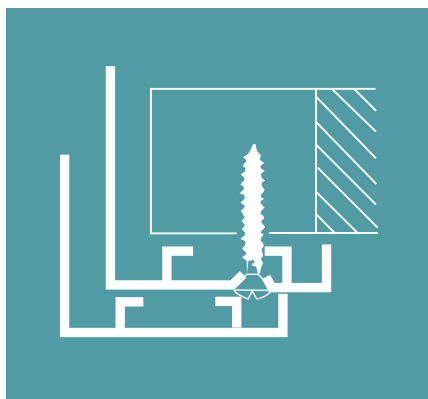


ECG-FD WITH DOUBLE FRAME*

TECNALCO model ECG-FD with Double Frame is a Fresh Air Egg Crate Grille with aluminium filter and opposed blade type volume control damper.

Overall grille depth = 47 + 36 mm

* The advantage of double frame provides easy removal of filter and operation of volume control damper.



FIXING DETAIL FOR DOUBLE FRAME GRILLE:

Is a concealed screw type fixing as shown in the figure.

N = Nominal size

A = Neck size of internal frame

a = Neck size of external frame

B = Over all size of the grille

where

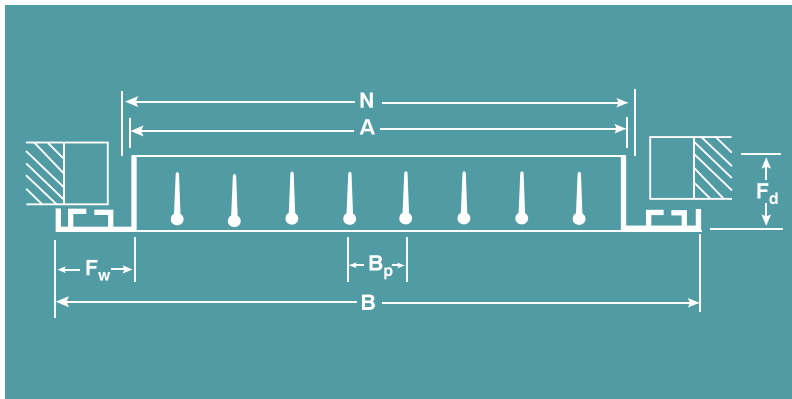
$N = L \times W$

$A = (L - 10) \times (W - 10) = (x \times y)$

$B = (x + 60) \times (y + 60) = (L + 50) \times (W + 50)$

$a = (x - 10) \times (y - 10)$

CONSTRUCTIONAL DETAILS AND DIMENSIONS



TECHNICAL DATA:

For Models RG types

N = Nominal size = L x W

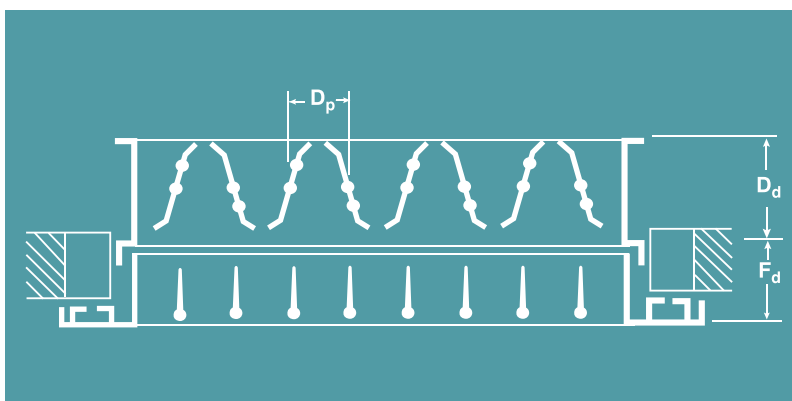
A = (L - 10) x (W - 10) = x x y

B = (x + 60) x (y + 60) =
(L + 50) x (W + 50)

Bp = Blade pitch = 19mm

Fd = Frame depth = 27 mm

Fw = Frame width = 30 mm

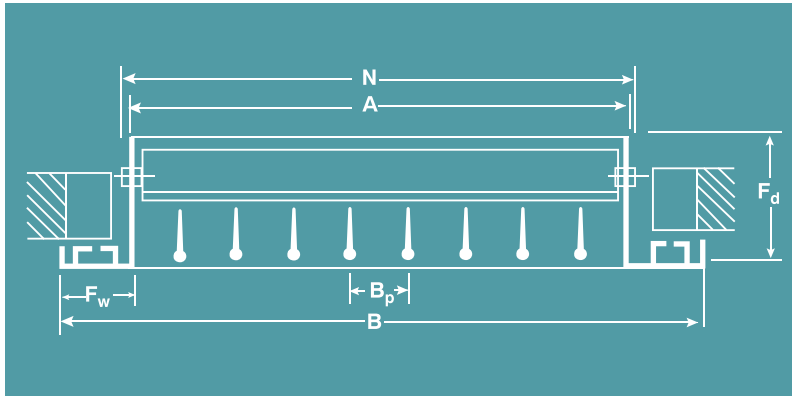


Dp = Damper pitch = 25 mm

Dd = Damper depth = 36 mm

Overall grille depth = 27 + 36 mm

○ Thickness of wooden frame required = 20 mm



For Models SG types

N = Nominal Size = L x W

A = (L - 10) x (W - 10) = x x y

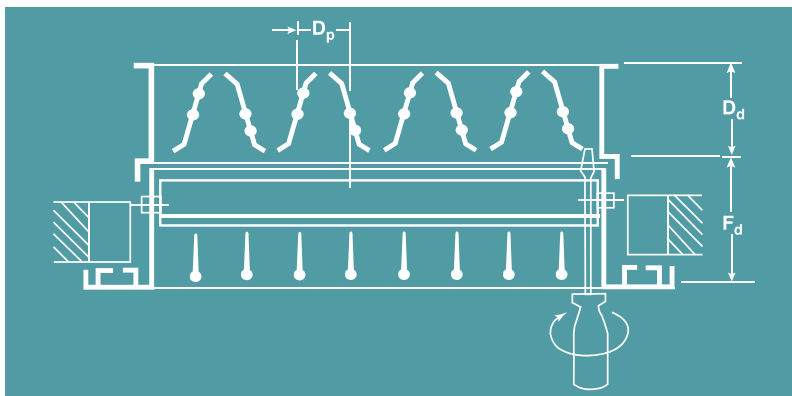
B = (x + 60) x (y + 60) =
(L + 50) x (W + 50)

Bph = Pitch of the horizontal blade = 19 mm

Bpv = Pitch of the vertical blade = 19 mm

Fd = Frame depth = 47mm

Fw = Frame width = 30mm



Dp = Damper pitch = 25 mm

Dd = Damper depth = 36 mm

Overall grilles depth = 47 + 36 mm

○ Thickness of wooden frame required = 20 mm

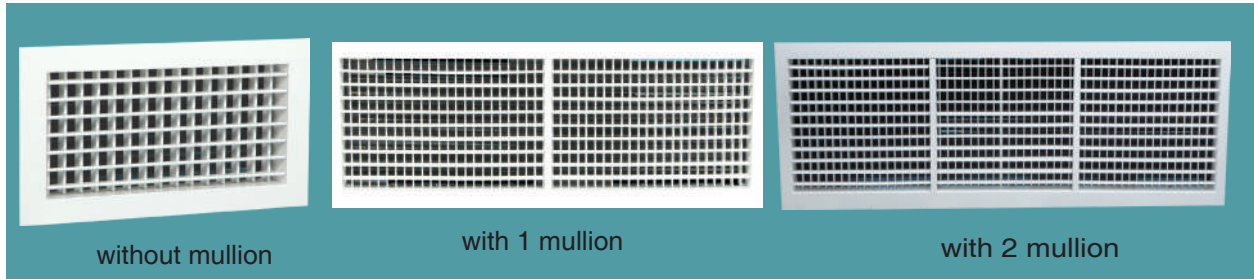
○ Operation of the Damper with a key (or) screw driver from the face of the grille, as shown in the figure.

CONSTRUCTIONAL DETAILS AND DIMENSIONS

Mullion Arrangements

L x W = Nominal size = Duct size

If	W or L ≤ 500 mm	→	without mullion
	W or L > 500 mm	→	1 mullion
	W or L > 1000 mm	→	2 mullions



STANDARD SIZES OF TECNALCO:

Aluminium Grilles Types: RG-H, H45, HD, VD, F, FD; SG-H, V, HD, VD; ECG, ECG-D, F; PG, PG-D

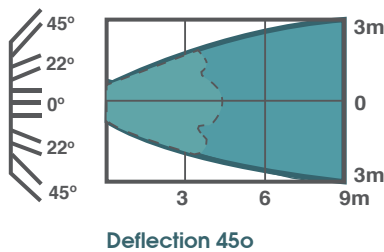
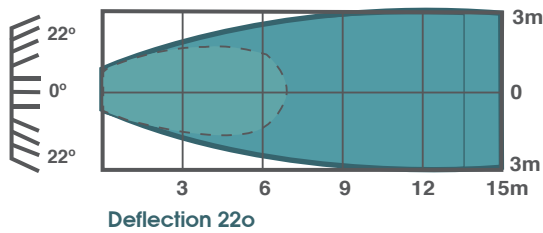
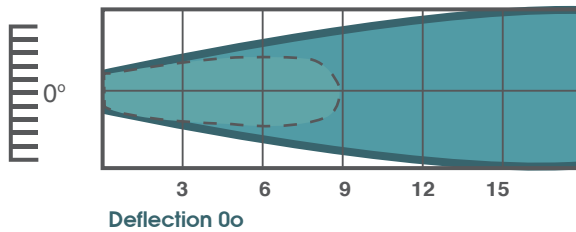
Standard Sizes L x W (mm).

150 x 100	550 x 150	1050 x 200	600 x 300	400 x 400	1150 x 450
200	600	1100	650	450	1200
250	650	1150	700	500	
300	700	1200	750	550	500 x 500
350	750		800	600	550
400	800	250 x 250	850	650	600
450	850	300	900	700	650
500	900	350	950	750	700
550	950	400	1000	800	750
600	1000	450	1050	850	800
650	1050	500	1100	900	850
700	1100	550	1150	950	900
750	1150	600	1200	1000	950
800	1200	650		1050	1000
850		700	350 x 350	1100	1050
900		750	400	1150	1100
950	200 x 200	800	450	1200	1150
1000	250	850	500		1200
1050	300	900	550		
1100	350	950	600	450 x 450	600 x 600
1150	400	1000	650	500	650
1200	450	1050	700	550	700
	500	1100	750	600	750
	550	1150	800	650	800
150 x 100	600	1200	850	700	850
200	650		900	750	900
250	700		950	800	950
300	750	300 x 300	1000	850	1000
350	800	350	1050	900	1050
400	850	400	1100	950	1100
450	900	450	1150	1000	1150
500	950	500	1200	1050	1200
550	1000	550		1100	

NOTE: The dark shaded portion indicates sizes available as standard.

ENGINEERING AND PERFORMANCE DATA

Deflection settings and air patterns.



Instruction for use of Drop Chart:

Find the total drop for an outlet of 0o deflection.

Discharge velocity = 3 m/s

Throw = 6 m

Temp diff. = 10o

Solution:

On the drop chart, lay a straight line connecting the values for velocity and throw, from examples above shown, by dark straight line.

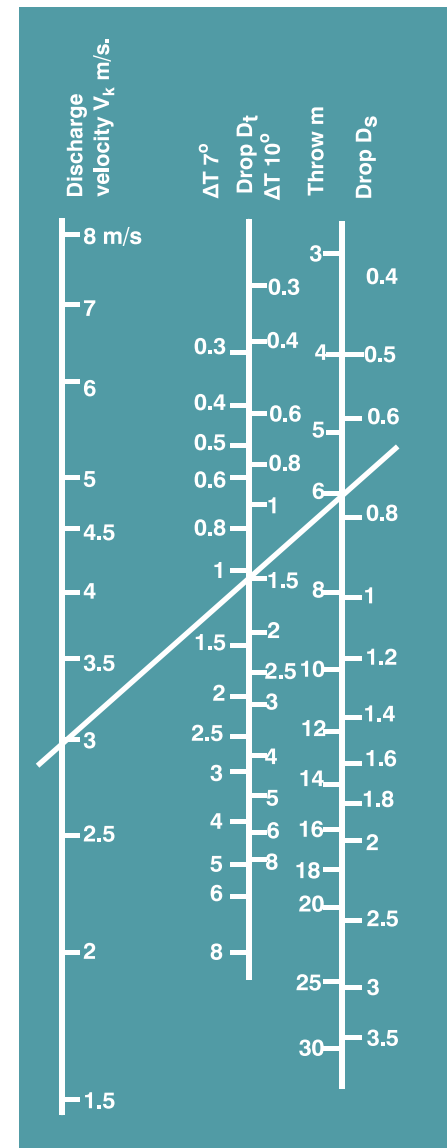
Total drop of the air stream is found by adding drop due to temp. Diff. (ΔT 10o Dt scale) and drop due to spread (Ds scale).

$$DT = Dt + Ds = 1.5 + 0.7 \text{ m} = 2.2 \text{ m.}$$

NOTES:

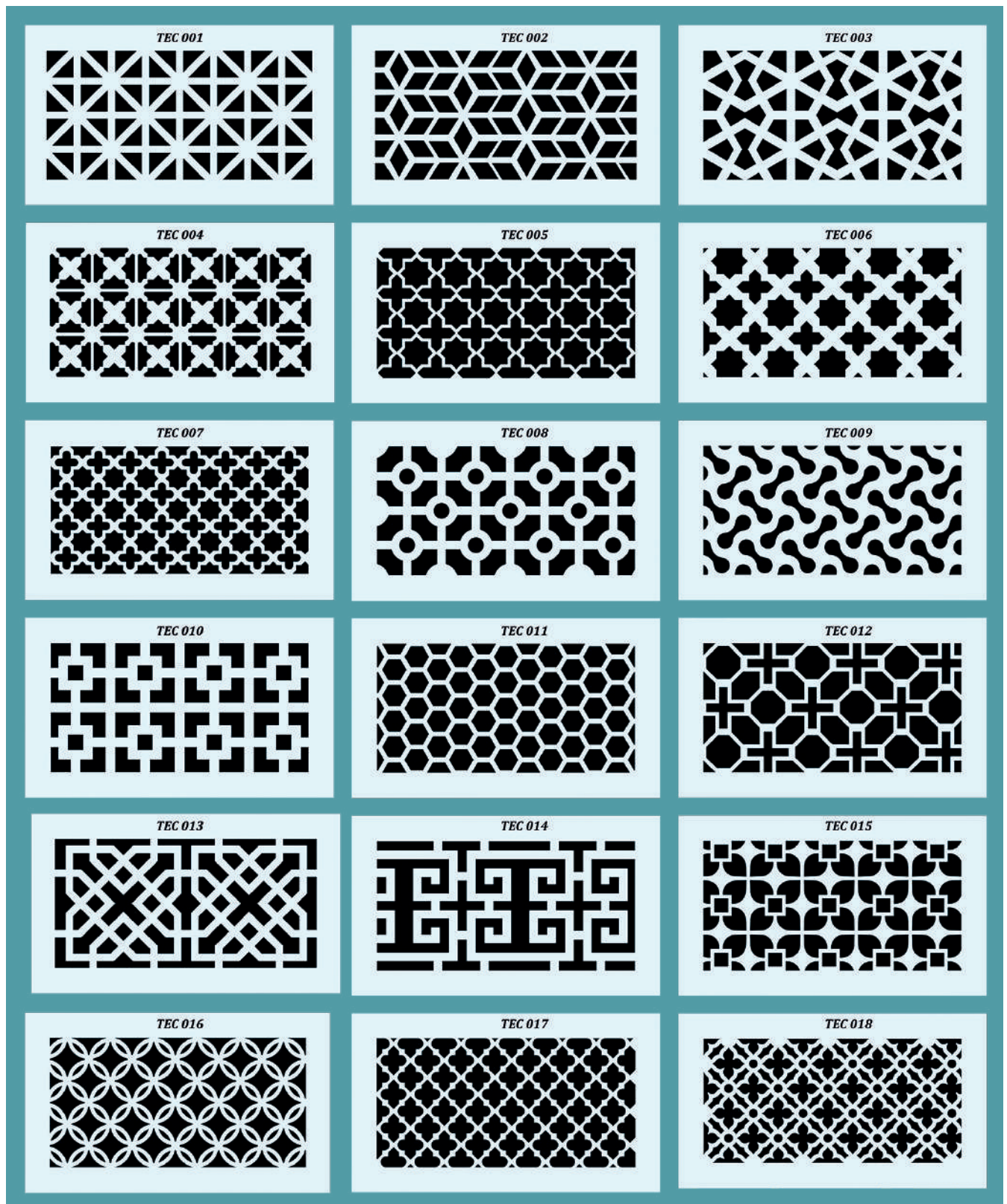
- Grilles with vertical blades and blade setting of 0o have the longest throw and the smallest spread.
- Changing the blade setting towards a longer angle will decrease the throw and increase the spread and induction. So a better mixing of supply air and room air will occur.
- With the double deflection grilles, the spread can be adjusted in the horizontal and vertical place.
- Blade setting over 45o is not recommended.

DROP CHART



Deflection	Throw	Vk	Pt	N.C.
22O	x 0.7	x 1.2	x 1.4	+ 3
45O	x 0.5	x 1.4	x 2	+ 4

DECORATIVE GRILLS



ENGINEERING AND PERFORMANCE DATA

Notes on Selection

- The throw is normally selected up to 3/4 of the room width.
- For a large air volume where the throw is more than 3/4 of the room width distributing the air stream over several outlets will reduce the throw.
- The drop of the selected outlet plus 1.8 met. is the minimum grille or register height from the floor as shown in the figs.
- From the selection diagrams the size of the grille can be selected taking into account the throw, the velocity, pressure loss and noise level of the grille.
- For supply grilles and registers, in addition to air flow, air throw and spread characteristics are the principle factors for selection.
- Variable deflection settings to satisfy the air distributing requirements can be obtained by individual adjustment of blades in TECNALCO supply grilles and registers.
- In order to obtain long throw and narrow air pattern use a deflection between 0° and 15° angle. For shorter throw and wide air pattern use up to 45° angle and even more under certain circumstances.
- Performance data shown in the selection charts on the following pages is based on double deflection grilles with horizontal blades set at 0° angle as illustrated
- Variation of throw in function of deflection setting is illustrated systematically for 22.5° angle and for 45° angle setting.
- If volume control damper is used, in conjunction of supply grille models, the effective area is reduced by approximately 5%. The effect is negligible on throw and temperature drop.

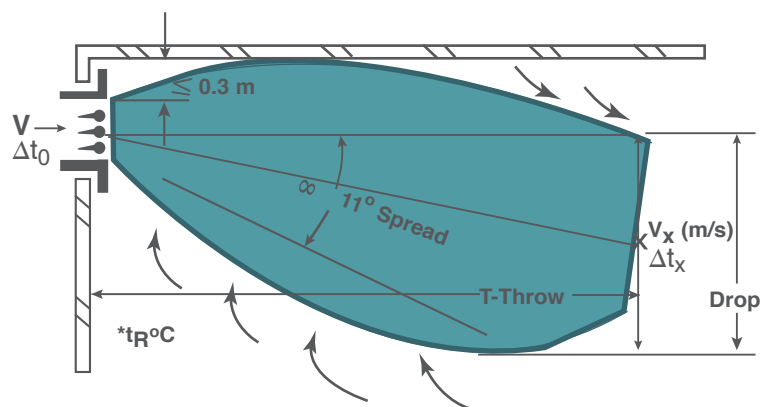
Nomenclature:

Where

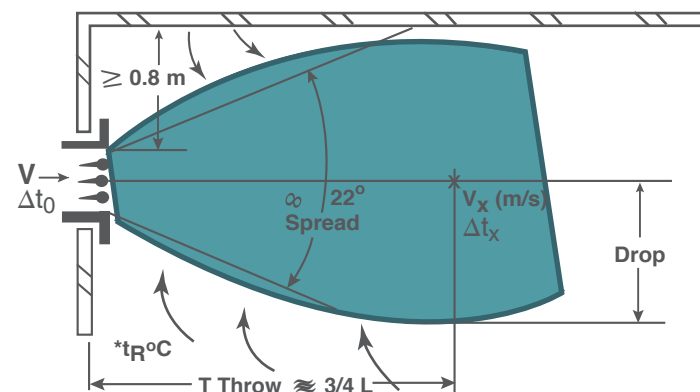
t_{RoC} = Room temperature

Δt_o = Supply air temperature differential

Type RG - with ceiling effect



Type RG - without ceiling effect



V = Volume of Air in m³/Sec

A_k = Effective free area of a grille in m²

V_k = Neck velocity of a grille in m / sec.

V_t = Terminal velocity in m /sec.

ΔP_t = Partical pressure in Pa

N.C. = Noise criteria

ENGINEERING AND PERFORMANCE DATA

When volume control damper is partially closed for balancing purposes or for final air flow control, in addition to pressure drop and sound corrections, the throw pattern will be reduced between 10 and 18% depending on the amount of throttling. Pressure drop will increase accordingly.

The sound level of a supply grille is in direct ratio to the velocity of the air pressure through it.

Air passing through a properly selected outlet grille will not add any appreciable noise to the sound level of the existing system.

In order for a new air conditioning system to perform to the designers' plans and specifications, it must be properly balanced to deliver the required amount of air through each diffuser.

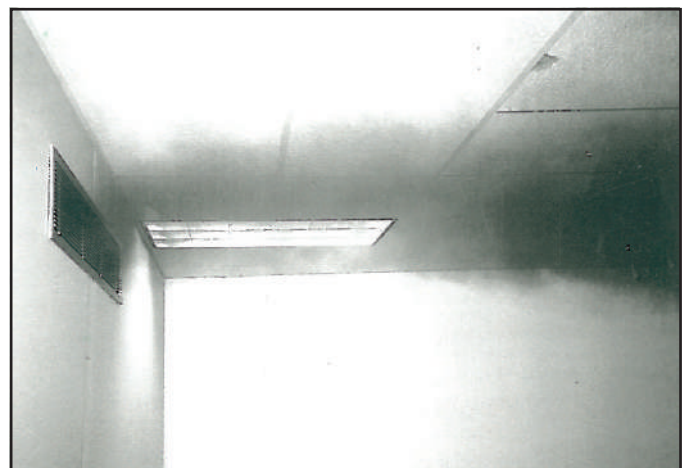
For vertically installed grilles and registers, the throw should reach approximately 3/4 of the distance from grille to the opposite wall.

Correct ceiling heights must be observed in order to prevent air stream to drop into the occupied zone which is generally about 1,8 m. above floor level.

Discharging of air pattern can be observed from the smoke test results



Model : RGHD 900 x 150 mm
Discharging of air with 0° blade deflection



Model : RGHD 900 x 250 mm
Discharging of air with 0° blade deflection

ENGINEERING AND PERFORMANCE DATA

Technical Data for TECNALCO Models RG & SG types

Selection Tables: For Area Factor Ak

L x W (mm)	*Ak
150 x 100	0.008
200	0.011
300	0.018
400	0.024
500	0.030
600	0.036
700	0.042
800	0.048
900	0.055
1000	0.060
1100	0.067
1200	0.073
150 x 150	0.013
200	0.017
300	0.027
400	0.036
500	0.045
600	0.055
700	0.064
800	0.074
900	0.084
1000	0.092
1100	0.102
1200	0.111
200 x 200	0.023
300	0.036
400	0.049
500	0.061
600	0.074
700	0.087
800	0.099
900	0.112
1000	0.124
1100	0.137
1200	0.150
250 x 250	0.038
300	0.047
400	0.063
500	0.079
600	0.095
700	0.112
800	0.129
900	0.145
1000	0.161
1100	0.178
1200	0.194

For models RG & SG types

○ K Factor

K = 0.6 ~ 0.65

Ak = K x total neck area
of the grille

*Ak = effective free area
Met2

ENGINEERING AND PERFORMANCE DATA

Selection Tables: For Area Factor A_k

For models RG & SG types

K Factor

$K = 0.6 \sim 0.65$

$A_k = K \times \text{Neck area of the grille}$

* A_k = effective free area Met2

Technical data for the other sizes available on request

NOTE:

For models ECG types

K factor

$K = 0.85 \sim 0.9$

For models PG types

$K = 0.3 \sim 0.35$

L x W (mm)	* A_k
300 x 300	0.056
400	0.076
500	0.095
600	0.114
700	0.134
800	0.154
900	0.174
1000	0.193
1100	0.213
1200	0.233
350 x 350	0.077
400	0.100
500	0.110
600	0.133
700	0.157
800	0.180
900	0.203
1000	0.225
1100	0.248
1200	0.271
400 x 400	0.103
500	0.128
600	0.155
700	0.182
800	0.209
900	0.236
1000	0.262
1100	0.289
1200	0.316
450 x 450	0.131
500	0.144
600	0.174
700	0.204
800	0.235
900	0.265
1000	0.293
1100	0.324
1200	0.354
500 x 500	0.160
600	0.193
700	0.227
800	0.260
900	0.294
1000	0.325
1100	0.359
1200	0.393
600 x 600	0.234
700	0.275
800	0.315
900	0.356
1000	0.394
1100	0.435
1200	0.476

ENGINEERING AND PERFORMANCE DATA

The tests were conducted, by the ETL Testing Laboratories, on the selected sizes of the grilles. The grilles had adjustable opposed blade dampers attached to their inlets. The dampers were set to full open position. The test results include Area factor, Throw pattern and sound power level.

Air volume q_v : Air volumes are given in Lt/sec.

Throw T : The throw values given in the tables are related to 00 blade divergence without ceiling effect.

Two Throw values are given for each q_v to the corresponding terminal velocities. V_t .

Min. Throw → 0.75 m/sec. V_t

Max. Throw → 0.50 m/sec. V_t

If the grille is near to the ceiling, these Throw values should be increased by a factor 1.4.

Discharge velocity V_k : The effective discharge velocities given are related to the corresponding A_k effective free areas.

N.C. values : Noise data is given in terms of N.C. levels. These have been derived from sound power level data (dB re 10-12 watts) with a 10 dB reduction for room effect.

Outlet size : Outlet sizes are mentioned on the head of each column with a range of sizes of equivalent A_k effective areas within a tolerance of $\pm 10\%$

SELECTION TABLES FOR TECNALCO MODELS RG & SG TYPES

L/S	List Sizes L x H (mm) Deflection	-200 x 100 0 Deg.		250 x 100 0 Deg.		300 x 100 200 x 150 0 Deg.		400 x 100 250 x 150 200 x 200 0 Deg.		450 x 100 300 x 150 0 Deg.	
25	Velocity m/s	2.22	15	1.74	15						
	Throw - m	1.55-2.2		1.3-2.0							
50	Velocity m/s	4.42	<20	3.46	<20	2.88	15	2.16	15	1.86	15
	Throw - m	3.0-4.5		2.6-4.0		2.4-3.6		2.1-3.1		1.9-2.9	
75	Velocity m/s	6.64	<20	5.20	<20	4.30	<20	3.26	<20	2.80	<20
	Throw - m	4.5-6.7		4.0-5.9		3.6-5.4		3.1-4.7		2.9-4.4	
100	Velocity m/s	8.86	20	6.94	20	5.74	<20	4.34	<20	3.72	<20
	Throw - m	6.0-9.0		5.3-8.0		4.8-7.2		4.1-6.3		3.9-5.8	
125	Velocity m/s	11.06	25	8.68	20	7.18	20	5.42	20	4.66	20
	Throw - m	7.5-11.2		6.8-10.2		6.0-9.0		5.2-8.0		4.8-7.2	
150	Velocity m/s			10.42	25	8.62	25	6.52	23	5.58	22
	Throw - m			8.0-12.0		7.2-10.8		6.3-9.4		5.8-8.7	
175	Velocity m/s					10.06	27	7.60	25	6.52	24
	Throw - m					8.4-12.6		7.3-11.0		6.8-10.2	
200	Velocity m/s							8.68	28	7.44	26
	Throw - m							8.4-12.5		7.7-11.6	
225	Velocity m/s							9.78	28	8.38	28
	Throw - m							9.4-14.1		8.7-13.1	
250	Velocity m/s							10.86	30	9.30	30
	Throw - m							10.4-15.7		9.7-14.5	
275	Velocity m/s									10.24	30
	Throw - m									10.7-16.0	

ENGINEERING AND PERFORMANCE DATA

SELECTION TABLES FOR TECNALCO MODELS RG & SG TYPES

L/S	List Sizes (mm)	500 x 100 250 x 200		700 x 100 450 x 150 250 x 250		750 x 100 500 x 150 350 x 200 300 x 250		950 x 100 600 x 150 450 x 200 350 x 250 300 x 300		1200 x 100 450 x 250 350 x 300	
		0 Deg.		0 Deg.		0 Deg.		0 Deg.		0 Deg.	
50	Velocity m/s Throw - m	1.68 1.8-2.8	15								
75	Velocity m/s Throw - m	2.52 2.8-4.1	15	1.84 2.4-3.5	15	1.66 2.2-3.3	15				
100	Velocity m/s Throw - m	3.38 3.7-5.5	<20	2.46 3.2-4.7	15	2.22 3.0-4.5	15	1.80 2.7-4.0	15		
125	Velocity m/s Throw - m	4.22 4.6-6.9	<20	3.08 3.9-5.9	<20	2.78 3.8-5.6	<20	2.24 3.4-5.0	15	1.78 3.0-4.5	15
150	Velocity m/s Throw - m	5.06 5.5-8.3	20	3.70 4.7-7.1	<20	3.34 4.5-6.8	<20	2.70 4.0-6.0	<20	2.14 3.6-5.4	15
175	Velocity m/s Throw - m	5.90 6.5-9.7	20	4.32 5.5-8.3	<20	3.88 5.3-7.8	<20	3.14 4.7-7.0	<20	2.50 4.2-6.2	<20
200	Velocity m/s Throw - m	6.74 7.4-11.0	22	4.94 6.3-9.4	20	4.44 6.0-9.0	20	3.60 4.1-8.1	<20	2.84 4.8-7.2	<20
225	Velocity m/s Throw - m	7.58 8.3-12.4	25	5.54 7.1-10.7	20	5.00 6.8-10.1	22	4.04 6.0-9.1	20	3.20 5.4-8.1	<20
250	Velocity m/s Throw - m	8.44 9.2-13.9	27	6.16 7.9-11.8	25	5.55 7.5-11.2	25	4.48 6.7-10.1	20	3.56 6.0-9.0	<20
275	Velocity m/s Throw - m	9.28 10.2-15.2	28	6.78 8.7-13.0	25	6.10 8.2-12.3	25	4.94 7.4-11.0	22	3.92 6.6-9.9	20
300	Velocity m/s Throw - m	10.12 11.1-16.5	30	7.40 9.4-14.2	28	6.66 9.0-13.4	28	5.38 8.0-12.1	25	4.28 7.2-10.8	20
325	Velocity m/s Throw - m			8.02 10.2-15.3	30	7.22 9.7-14.6	30	5.84 8.7-13.1	25	4.62 7.7-11.6	20
350	Velocity m/s Throw - m			8.64 11.0-16.5	30	7.78 10.4-15.7	30	6.28 9.4-14.1	28	4.98 8.4-12.6	22
375	Velocity m/s Throw - m			9.24 11.8-17.7	35	8.32 11.2-16.8	33	6.74 10.1-15.1	30	5.34 9.0-13.5	22
400	Velocity m/s Throw - m			9.86 12.6-18.9	35	8.88 11.9-18.0	35	7.18 10.7-16.1	30	5.70 9.6-14.3	25
425	Velocity m/s Throw - m			10.48 13.4-20.1	40	9.44 12.7-19.0	35	7.62 11.4-17.1	33	6.06 10.2-15.3	25
450	Velocity m/s Throw - m					10.00 13.5-20.2	40	8.08 12.1-18.1	35	6.40 10.8-16.1	28
475	Velocity m/s Throw - m							8.52 12.8-19.1	35	6.76 11.4-17.0	30
500	Velocity m/s Throw - m							8.98 13.4-20.2	38	7.12 12.0-18.0	30
550	Velocity m/s Throw - m									7.84 13.2-19.7	35

ENGINEERING AND PERFORMANCE DATA

SELECTION TABLES FOR TECNALCO MODELS RG & SG TYPES

L/S	List Sizes L x H (mm)	900 x 150		1000 x 150		900 x 200		1200 x 200		1000 x 300	
		700 x 200		600 x 250		700 x 250		800 x 300		900 x 350	
		550 x 250		500 x 300		600 x 300		700 x 350		800 x 400	
		450 x 300		400 x 350		500 x 350		600 x 400		700 x 450	
Deflection	0 Deg.	0 Deg.	0 Deg.	0 Deg.	0 Deg.	0 Deg.	0 Deg.	0 Deg.	0 Deg.	0 Deg.	0 Deg.
175	Velocity m/s Throw - m	2.04 3.8-5.7	15	1.88 3.6-5.4	15	1.56 3.3-4.9	15				
200	Velocity m/s Throw - m	2.34 4.3-6.5	15	2.16 4.2-6.2	15	1.78 3.8-5.7	15				
225	Velocity m/s Throw - m	2.62 4.9-7.3	15	2.42 4.7-7.0	15	2.00 4.3-6.4	15	1.80 2.7-4.0	15		
250	Velocity m/s Throw - m	2.92 5.4-8.2	<20	2.70 5.3-7.8	<20	2.22 4.8-7.1	15	1.62 4.0-6.1	15	1.78 3.0-4.5	<15
275	Velocity m/s Throw - m	3.20 6.0-8.9	<20	2.96 5.8-8.6	<20	2.44 5.2-7.8	<20	1.78 4.5-6.7	15	2.14 3.6-5.4	<15
300	Velocity m/s Throw - m	3.50 6.5-9.7	<20	3.24 6.2-9.4	<20	2.66 5.7-8.5	<20	1.94 4.8-7.3	15	1.50 4.3-6.4	15
325	Velocity m/s Throw - m	3.78 7.0-10.5	<20	3.50 6.7-10.2	<20	2.88 6.1-9.2	<20	2.10 5.3-7.9	<20	1.62 4.5-6.9	15
350	Velocity m/s Throw - m	4.08 7.6-11.4	20	3.78 7.3-10.9	20	3.10 6.6-9.9	<20	2.26 5.7-8.4	<20	1.76 5.0-7.5	15
375	Velocity m/s Throw - m	4.36 8.1-12.1	20	4.04 7.8-11.7	20	3.32 7.1-10.6	20	2.42 6.0-9.1	<20	1.88 5.3-7.9	<20
400	Velocity m/s Throw - m	4.66 8.6-13-0	22	4.32 8.3-12.6	20	3.54 7.5-11.3	20	2.58 6.5-9.7	<20	2.00 5.7-8.4	<20
425	Velocity m/s Throw - m	4.96 9.2-13.8	22	4.58 8.9-13.3	22	3.76 8.0-12.0	20	2.74 6.8-10.3	20	2.12 6.0-9.0	<20
450	Velocity m/s Throw - m	5.24 9.7-14.6	25	4.86 9.4-14.1	22	3.98 8.4-12.7	22	2.90 7.2-10.9	20	2.26 6.4-9.6	<20
475	Velocity m/s Throw - m	5.54 10.3-15.4	25	5.12 9.9-14.8	25	4.20 8.9-13.4	22	3.06 7.7-11.5	20	2.38 6.7-10.1	20
500	Velocity m/s Throw - m	5.82 10.8-16.2	28	5.40 10.4-15.7	25	4.44 9.4-14.2	22	3.22 8.0-12.0	20	2.50 7.1-10	20
550	Velocity m/s Throw - m	6.40 11.9-17.8	28	5.92 11.4-17.2	28	4.88 10.4-15.6	25	3.54 8.9-13.3	22	2.76 7.8-11.7	20
600	Velocity m/s Throw - m	7.00 13.0-19.5	30	6.46 12.5-18.7	30	5.32 11.4-17.0	25	3.86 9.7-14.5	22	3.00 8.5-12.8	20
650	Velocity m/s Throw - m	7.58 14.1-21.1	30	7.00 13.6-20.3	30	5.76 12.3-18.4	28	4.20 10.4-15.8	25	3.26 9.2-13.8	22
700	Velocity m/s Throw - m	8.16 15.2-22.7	35	7.54 14.6-21.9	33	6.20 13.2-19.8	30	4.52 11.3-17.0	25	3.50 9.9-14.9	22
750	Velocity m/s Throw - m	8.74 16.3-24.4	35	8.08 15.6-23.4	35	6.64 14.1-21.2	30	4.84 12.0-18.2	28	3.76 10.7-16.0	22
800	Velocity m/s Throw - m			8.62 16.7-25.0	37	7.10 15.1-22.7	33	5.16 12.9-19.3	28	4.00 11.3-17.0	25

ENGINEERING AND PERFORMANCE DATA

SELECTION TABLES FOR TECNALCO MODELS RG & SG TYPES

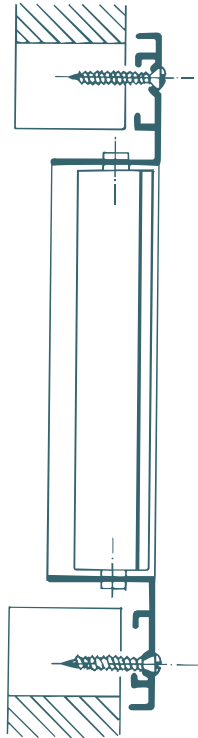
L/S	Deflection	1100 x 350		1200 x 300		1200 x 400		1200 x 500		1200 x 600	
		List Sizes L x H (mm)	0 Deg.	0 Deg.	0 Deg.	0 Deg.	0 Deg.	0 Deg.	0 Deg.	0 Deg.	0 Deg.
400	Velocity m/s Throw - m	1.66 5.2-7.7	<15								
425	Velocity m/s Throw - m	1.78 5.5-8.3	<15	1.52 5.1-7.7	<15						
450	Velocity m/s Throw - m	1.88 5.8-8.7	<15	1.62 5.4-8.2	<15						
475	Velocity m/s Throw - m	1.98 6.2-9.2	15	1.70 5.7-8.5	<15						
500	Velocity m/s Throw - m	2.08 6.5-9.7	15	1.80 6.0-9.0	<15	1.56 5.6-8.4	<15				
550	Velocity m/s Throw - m	2.30 7.5-11.2	15	1.98 6.6-9.9	15	1.70 6.1-9.2	<15				
600	Velocity m/s Throw - m	2.50 7.7-11.6	15	2.16 7.2-10.8	15	1.86 6.7-10.0	<15	1.52 6.0-9.0	<15		
650	Velocity m/s Throw - m	2.70 8.4-12.6	<20	2.34 7.8-11.7	15	2.02 7.2-10.9	<15	1.64 6.5-9.8	<15		
700	Velocity m/s Throw - m	2.92 9.1-13.6	<20	2.52 8.4-12.6	15	2.18 7.8-11.7	15	1.76 7.0-10.5	<15		
750	Velocity m/s Throw - m	3.12 9.7-14.6	<20	2.70 9.0-13.6	<20	2.34 8.4-12.6	15	1.90 7.6-11.4	<15	1.58 6.9-10.4	<15
800	Velocity m/s Throw - m	3.34 10.4-15.5	<20	2.88 9.7-14.5	<20	2.48 8.9-13.3	15	2.02 8.0-12.1	15	1.68 7.3-11.0	<15
850	Velocity m/s Throw - m	3.54 11.0-16.5	20	3.04 10.2-15.3	<20	2.64 9.4-14.2	15	2.14 8.5-12.8	15	1.78 7.8-11.6	<15
900	Velocity m/s Throw - m	3.76 11.6-17.5	20	3.22 10.8-16.2	<20	2.80 10.1-15.1	<20	2.28 9.1-13.6	15	1.90 8.3-12.5	15
950	Velocity m/s Throw - m	3.96 12.2-18.5	22	3.40 11.4-17.0	20	2.96 10.7-15.9	<20	2.40 9.5-14.3	15	2.00 8.7-13.1	15
1000	Velocity m/s Throw - m	4.18 13.0-19.5	22	3.58 12.0-18.0	20	3.10 11.1-16.7	<20	2.52 10.1-15.0	<20	2.10 9.2-13.8	15
1050	Velocity m/s Throw - m	4.38 13.6-20.4	22	3.76 12.6-18.9	20	3.26 11.7-17.5	<20	2.66 10.6-15.9	<20	2.20 9.7-14.4	15
1100	Velocity m/s Throw - m	4.58 14.2-21.3	25	3.94 13.2-19.8	22	3.42 12.3-18.4	<20	2.78 11.1-16.6	<20	2.32 10.2-15.2	<20
1150	Velocity m/s Throw - m	4.80 14.9-22.4	25	4.12 13.8-20.7	22	3.58 12.9-19.3	<20	2.90 11.6-17.3	<20	2.42 10.6-15.9	<20
1200	Velocity m/s Throw - m	5.00 15.5-23.3	25	4.30 14.4-21.6	22	3.74 13.4-20.2	22	3.04 12.1-18.2	20	2.52 10.9-16.4	<20
1250	Velocity m/s Throw - m	5.22 16.2-24.3	25	4.48 15.0-22.5	25	3.88 13.9-20.9	22	3.16 12.6-18.9	20	2.62 11.4-17.2	20

Grilles and Registers Fixing Details

Type 'S' fixing:

A counter-sunk screw type fixing

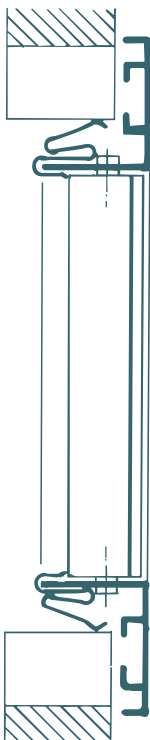
- Recommended for sidewall, ceiling and sill applications.



Type 'C' fixing:

Concealed type fixing with spring clip

- Recommended for sidewall and sill applications.



GRILLES AND REGISTERS FIXING DETAILS

Type 'C' fixing:

Concealed type fixing with screw lever

- Recommended for sidewall, ceiling and sill applications.

Note:

- Standard supply is Type 'S' fixing.

Available Finishes:

- Natural anodised aluminium
- Powder coated colour finish.
(Mention RAL colour number)
- Mill finish aluminium

○ Available Finish for the accessories

Mill finish aluminium

Powder coated in black matt finish

○ Ordering Data:

specify

1. TECNALCO type and model
2. Nominal size - L x W
3. Quantity
4. Type of fixing
5. Surface finish
6. Remarks

Example:					
1.	2.	3.	4.	5.	6.
SGHD	900 x 500 mm	50 nos.	Type 'S' fixing	Natural aluminium finish	D → Damper Mill finish



TECNALCO

Quality Air Distribution Since 1981

ALUMINIUM TECHNICAL ENGINEERING FACTORY

Manufacturer of high quality Air distribution devices and Duct accessories

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